Global Institute for Water Security
USASK
SCHOOL OF ENVIRONMENT AND SUSTAINABILITY
UNIVERSITY OF SASKATCHEWAN

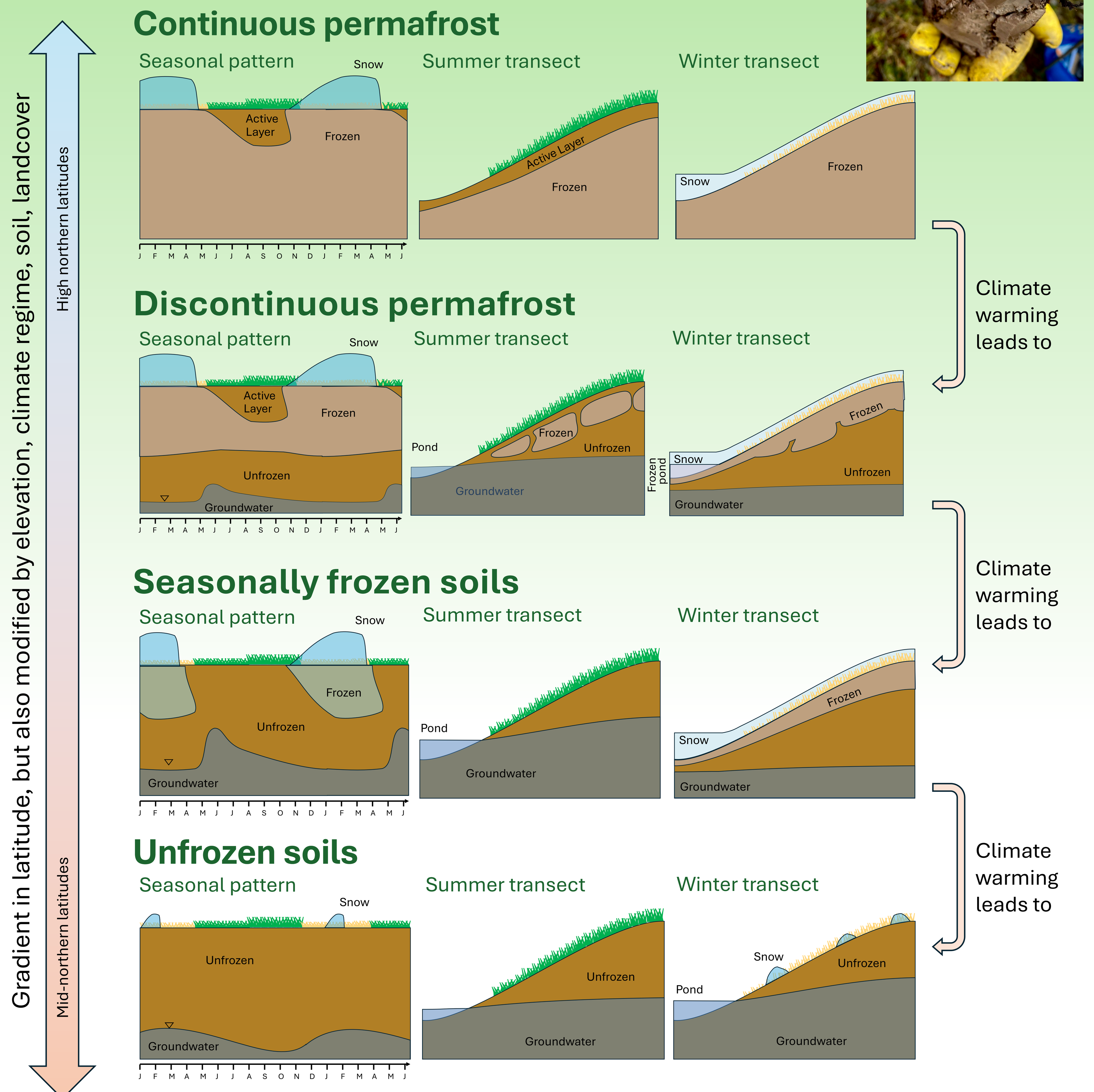
# Energy conservative solutions for coupled heat-mass transport in frozen soils

Andrew Ireson[1], Alana Muenchrath[1], Christopher Spence[2]
1. University of Saskatchewan, Saskatoon, Canada. 2. Environment and Climate Change Canada, Saskatoon, Canada

Funding was provided by
Environment and Climate Change Canada
Environnement et Changement climatique Canada

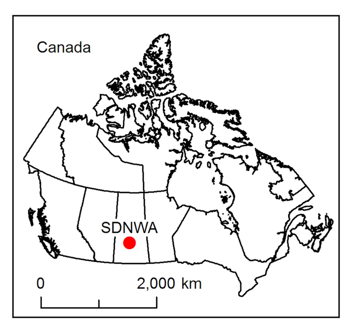
## The hydrological importance of frozen soils

Frozen soils are a critical component of cold regions hydrology. The presence of ice in soil pore space, which is dynamic in space and time, controls hydraulic conductivity, hydrological pathways, and infiltration-runoff partitioning. We developed a flexible modelling framework to allow us to explore and test our understanding of frozen soil hydrological processes.




## Canadian prairies

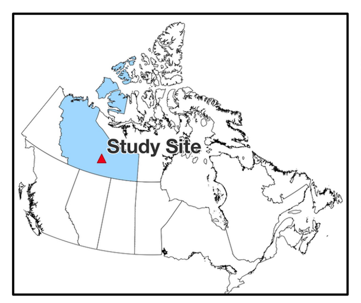
Snowmelt infiltration/runoff partitioning is a critical process. Infiltration is needed for crops, while runoff can cause flooding. Hydrological models often fail at accurately simulating streamflow due to the complex physics of infiltration in frozen soils.




Treaty 6 territory, the homeland of the the Plains and Wood Cree, Assiniboine, Dene, Nakoda, and Saulteaux First Nations and Métis people.

## Canadian shield

Under climate warming, permafrost in soil-filled hillslopes is susceptible to active layer thickening and talik development, changing hydrological pathways and solute export.




Treaty 11 encompasses the territories of Chief Drygeese, the Yellowknives Dene, and Tlicho First Nations, the North Slave Métis and Inuit people.

soilice

Ireson, A.M. (2026). amireson/soilice: v1.0 (v1.0). Zenodo. doi:10.5281/zenodo.19832264

## The soilice model

soilice is an open-source, Python-based **1D coupled heat-mass transport solver** for variably saturated/variably frozen soils. It is simple to install and run, with modular, readable code that can be modified to test alternative hypotheses about frozen soil processes. A *SciPy* ODE solver gives excellent mass and energy conservation (except for unphysical parameter combinations, e,g, that cause ice to flow). The *Numba* just-in-time compiler means the code is as efficient as compiled languages. Existing benchmarks are reproduced, but these do not rigorously test the coupling of flow and heat transport. Here we demonstrate coupled behavior with infiltration simulations.

Terminal/Powershell command to install the model using *pip*:

```
~ pip install git+https://git@github.com/amireson/soilice.git
```

Python script that sets up soilice to run an infiltration scenario

```python
import numpy as np                    # import numpy
from soilice import model             # type: ignore # import soilice

sim=model()                           # instantiate model
sim.opts={'simulateFlow': 1.0,        # Solve flow equations
        'gravity': 1.0,               # Vertical profile
        'infiltration': 1.0,          # Potential infiltration BC
        'cryoK': 0.0,                 # Use impedance model for K
        'cryoGradient': 0.0,          # Don't use cryosuction gradient
        'freeDrainage': 1.0,          # Free drainage lower BC
        'withadv': 1.0,               # Include advection
        'simulateTransport': 1.0,     # Solve heat transport equations
        'conductionTop': 1.0,         # Conductive heat flux at upper BC
        'conductionBot': 0.0}         # No conduction on lower BC

dz, zMax = 0.001, 2.0                 # Space grid
sim.zGrid(np.arange(0,zMax+dz,dz))
sim.tGrid(0,20,0.01)                  # Time grid
sim.readPars()                        # Load parameters from file

sim.setBCs(qI=0.05,TTop=0.)           # Assign boundary conditions
sim.setICs(psi0=-5,T0=-1)             # Assign initial conditions
out=sim.run()                         # Run model and return all output

import matplotlib.pyplot as pl        # import pyplot
pl.plot(out.T.T,out.z)                # plot temperature profiles
pl.show()
```

## Governing equations

**Conservation equations:**

$$\frac{\partial m}{\partial t} = \frac{\partial m_l}{\partial t} + \frac{\partial m_i}{\partial t} = -\rho_l \nabla \cdot \mathbf{q}_l \qquad \frac{\partial u}{\partial t} = -\nabla \cdot \mathbf{j}$$

**Equation of state:**

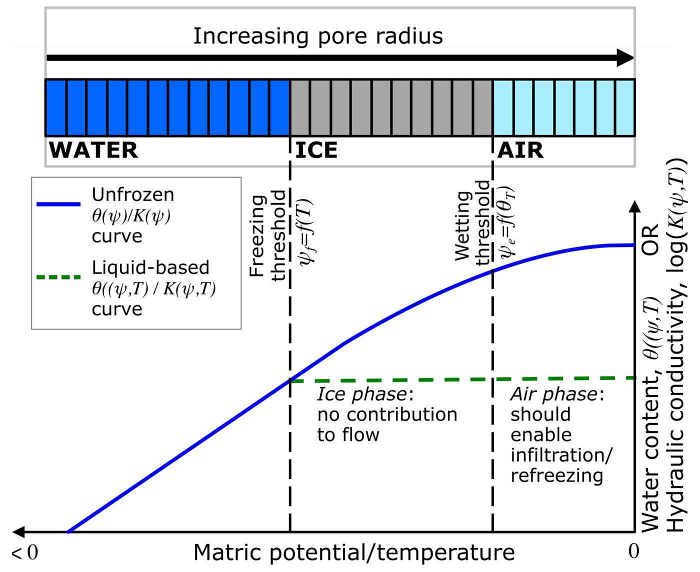
$$u = (m_i c_{pi} + m_l c_{pl} + m_s c_{ps})T - \lambda_f m_i$$

**Combined:**

$$c_{pb}\frac{\partial T}{\partial t} + (c_{pi}T - \lambda_f)\frac{\partial m_i}{\partial t} + c_{pl}T\frac{\partial m_l}{\partial t} = -\nabla \cdot \mathbf{j}$$

**Closure relations:**

$$\frac{dm_l}{dt} = \begin{cases} \rho_l C(\psi_e)\dfrac{d\psi_e}{dt}, & \psi_e \le \psi_f \\ \rho_l F'(T)\dfrac{dT}{dt}, & \psi_e > \psi_f \end{cases} \quad \text{and} \quad \frac{dm_i}{dt} = \begin{cases} 0, & \psi_e \le \psi_f \\ \rho_l C(\psi_e)\dfrac{d\psi_e}{dt} - \rho_l F'(T)\dfrac{dT}{dt}, & \psi_e > \psi_f \end{cases}$$




**Governing PDEs:**

Pressure-form of mass balance:

$$\frac{\partial \psi_e}{\partial t} = \frac{-\rho_l \nabla \cdot \mathbf{q}_l}{C(\psi_e)}$$

Temperature-form of unfrozen energy balance:

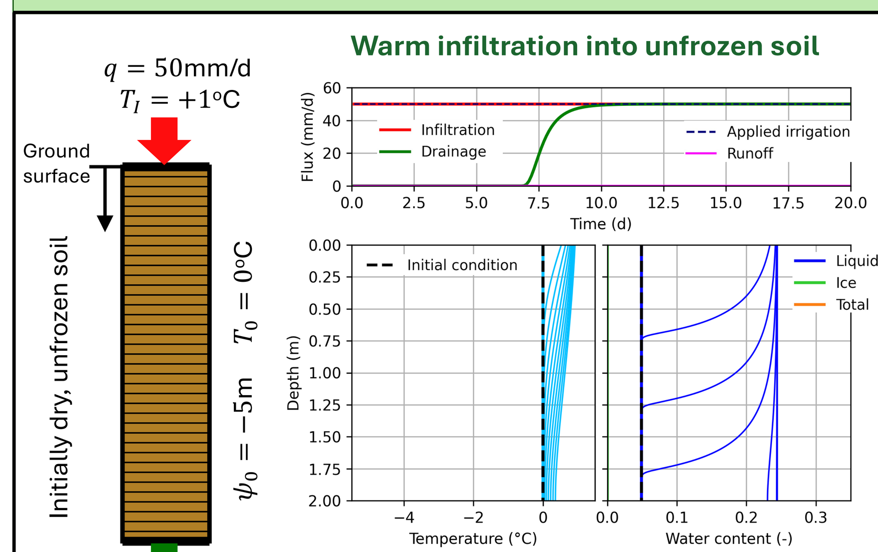
$$\frac{\partial T}{\partial t} = \frac{-\nabla \cdot \mathbf{j} + c_{pl}T\rho_l \nabla \cdot \mathbf{q}_l}{c_{pb}}$$

Temperature-form of frozen energy balance:

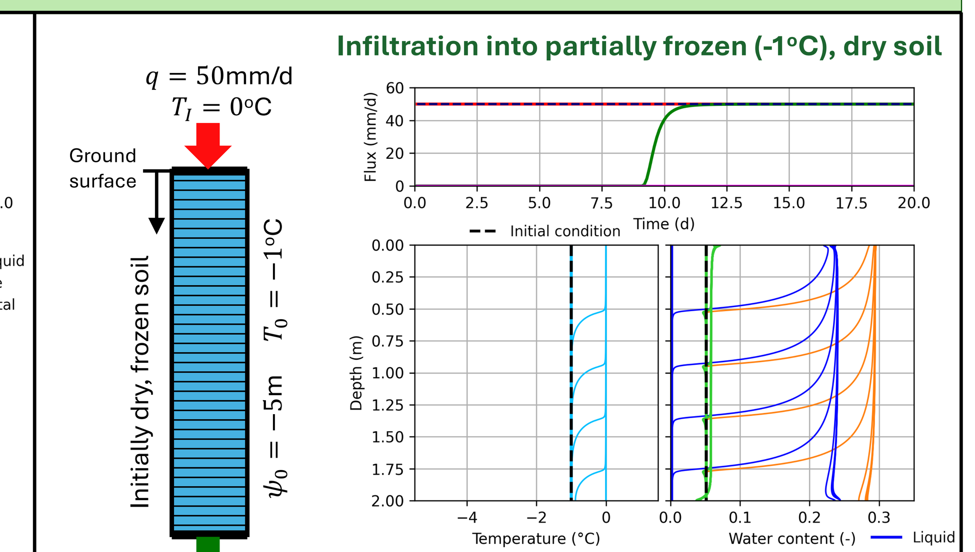
$$\frac{\partial T}{\partial t} = \frac{-\nabla \cdot \mathbf{j} + (c_{pi}T - \lambda_f)\rho_l \nabla \cdot \mathbf{q}_l}{c_{pb} + \rho_l F'(T)\left((c_{pl} - c_{pi})T + \lambda_f\right)}$$

## Infiltration in seasonally frozen soils

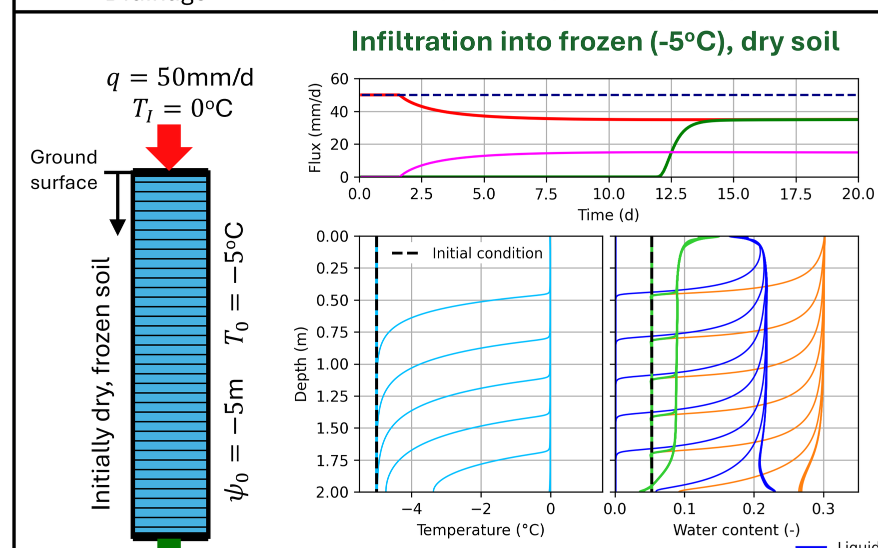
Snowmelt in the prairies occurs while the soils are frozen to depths > 1m, yet a surprisingly large portion of snowmelt infiltrates into frozen soils. Infiltration depends on the liquid/ice/air content at the time of melt – saturated and frozen soils are impermeable; dry frozen soils contain air-filled pores that permit infiltration. Here, using different initial conditions, we show how soilice reproduces this behavior. The infiltration flux is given by $q_T = \min\left(q_I, -K\left(\min(\psi_0, 0).\frac{2}{dz} - 1\right)\right)$



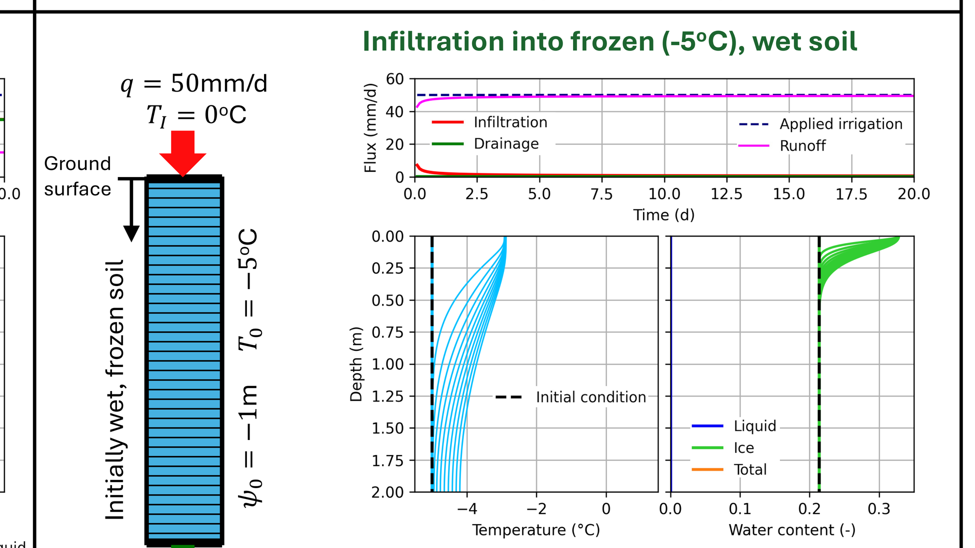

**Outcome:** 100% infiltration, no freezing, drainage breakthrough in 8 d.




**Outcome:** 100% infiltration, ice/liquid coexist, drainage breakthrough in 10 d.



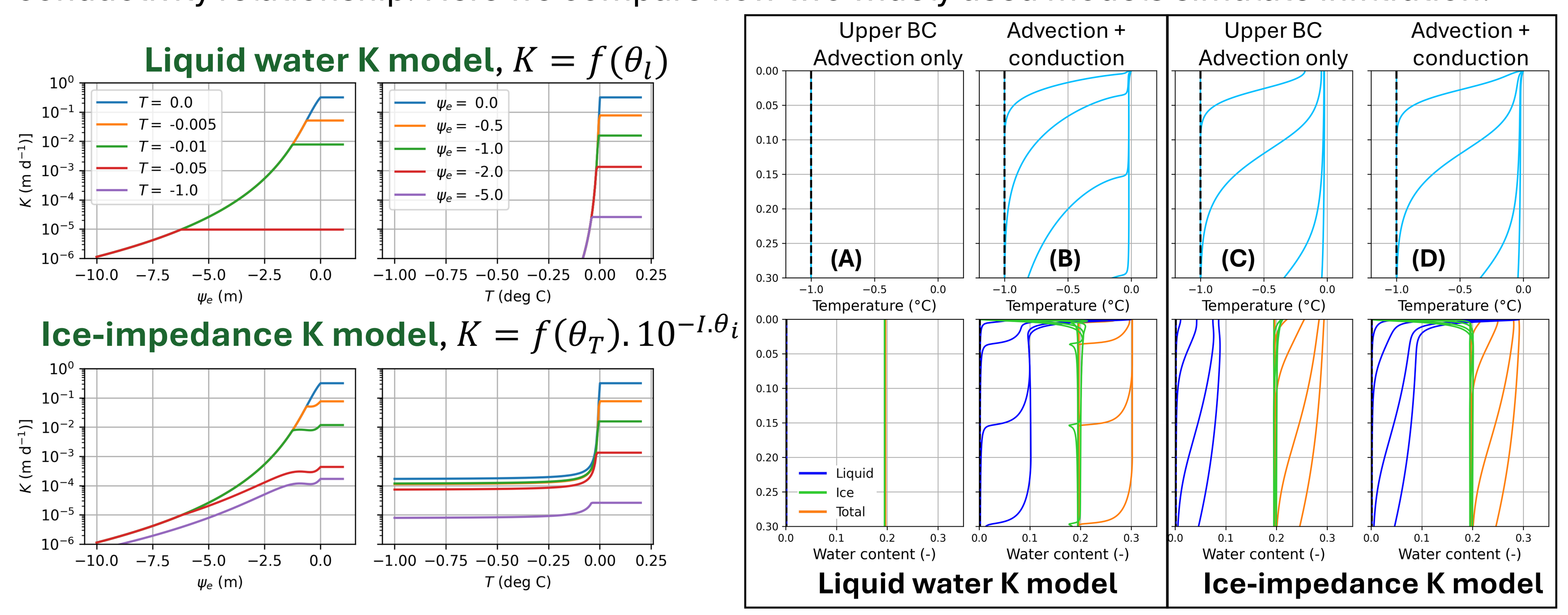

**Outcome:** 74% infiltration, 26% runoff, ice/liquid coexist, breakthrough in 13 d.




**Outcome:** 2% infiltration, 98% runoff, high ice content at surface, no breakthrough, minimal warming

## The importance of hydraulic conductivity, $K(\psi, T)$

The single most important control on the flow and heat transport in frozen soils is the hydraulic conductivity relationship. Here we compare how two widely used models simulate infiltration.




The ice-impedance K model allows for higher $K$ during frozen conditions, leading to higher infiltration rates and deeper wetting fronts (C/D). The liquid water K model has very low $K$ at -1°C, restricting infiltration (A) or producing sharp wetting fronts (B). The effect of warming the ground surface by conduction is small in the ice-impedance K model (C/D), as heat enters the profile by advection. In the liquid water K model, without surface warming by conduction, there is no infiltration or advected heat (A). The impedance model is more realistic for simulating infiltration, but there are two problems: *i)* the $K(\psi, T)$ relationship can be non-monotonic; and *ii)* higher $K$ in frozen conditions causes "ice flow", leading to energy balance errors.